

화학수학

열아홉 번째 수업

확률과 통계



Group (군)

Collection of members
with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the **product** AB is also a member of the group
2. The group must contain the **identity** E such that

$$AE = EA = A \text{ (for every member } A\text{)}$$

3. The **inverse** of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

4. The **associative law** holds: $A(BC) = (AB)C$

Point Group (점군)

Formed by a set of point symmetry operators
belonging to a symmetrical object

Representation of a Point Group

Symmetry operator $\leftrightarrow$ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

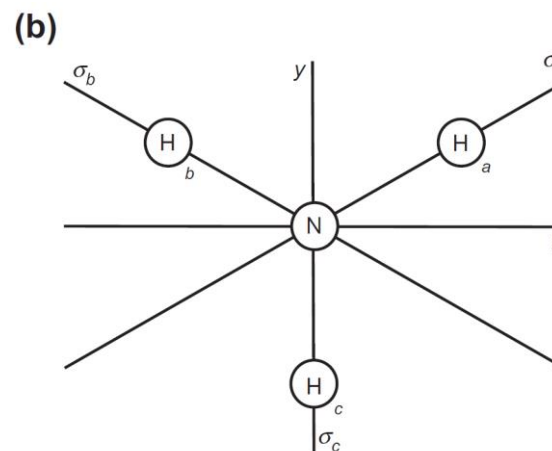
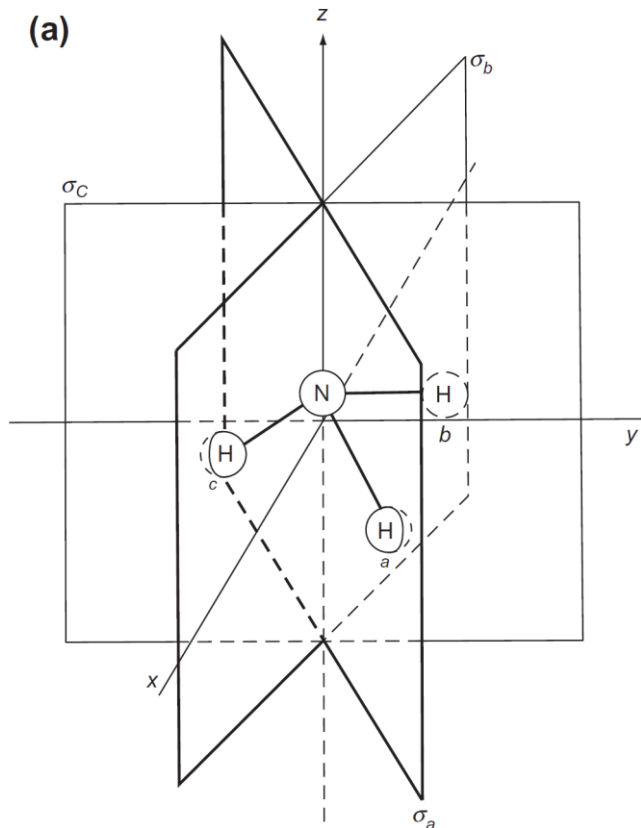
How to Find the Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}y_1 + a_{13}z_1 = x_2 \\ a_{21}x_1 + a_{22}y_1 + a_{23}z_1 = y_2 \\ a_{31}x_1 + a_{32}y_1 + a_{33}z_1 = z_2 \end{cases}$$

From the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) ,
determine a_{ij}

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

Similarity Transformation (답음 변환)

- Two different representations **P** and **Q** in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

when **X** is some square matrix

- Since $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$,

$$\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \text{tr}(\mathbf{XX}^{-1}\mathbf{Q}) = \text{tr}(\mathbf{Q})$$

Character (지표)

- **Character** of a symmetry operator = **trace** of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Character of } \hat{C}_3 = \text{tr } \mathbf{A} = (-1/2) + (-1/2) + 1 = 0$$

Reducible Representation (가약 표현)

All matrices are **block-diagonal**, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{F}$$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal
by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as **irrep**

Character Table (지표표)

- Using group theory theorems, we can find all possible irreps and collect their characters → **character table**
- e.g.* C_{3v} group

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

conventional notation for irreps

Today's Class

- Errors in measurement
- Probability theory
 - Discrete probability distributions
 - Continuous probability distributions
 - Examples

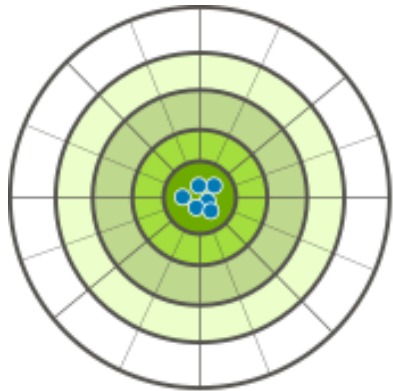
15. Probability, Statistics, and Experimental Errors

(15장 확률, 통계, 실험오차)

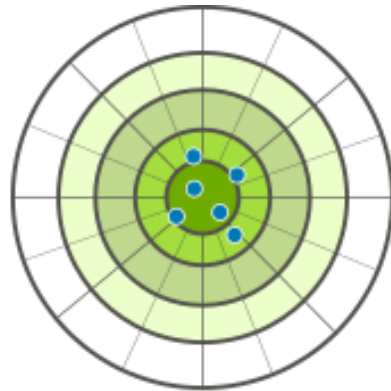
Errors in Measurement

- **Systematic errors** (계통오차)
 - same direction and same magnitude for repeated measurements
 - due to limitations in the apparatus
- **Random errors** (우연오차)
 - random direction and magnitude
 - due to limitations in the apparatus, limitations in the techniques, unavoidable human error

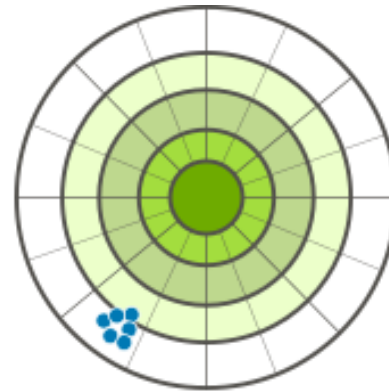
Precision (정밀도) and Accuracy (정확도)



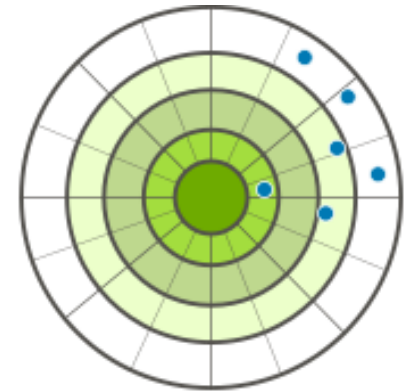
accurate
precise



accurate
not precise



not accurate
precise



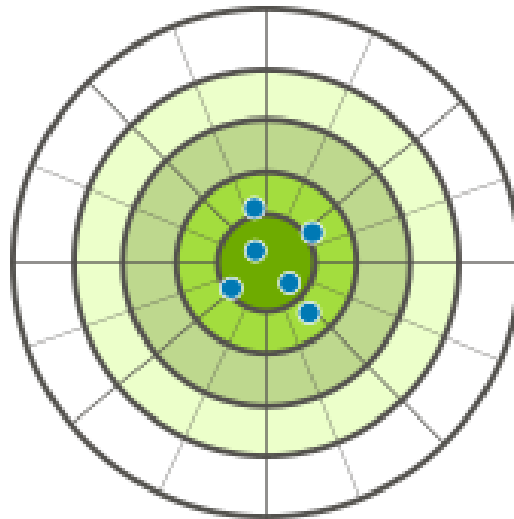
not accurate
not precise

random errors

systematic errors

Repeated Measurements

sampling from a population of imaginary measurements with random errors



(Image: <https://openclipart.org/detail/191649/accuracy-vs-precision>)

Probability Theory

study of random events

- **Classical systems:** unknown information
- **Quantum systems:** intrinsic randomness
- **Population:** large set of objects or numbers
 - Classical systems: imaginary set of infinitely many repetitions of a given measurement
 - Quantum systems: imaginary set of infinitely many replicas of the physical system

Probability Distribution (확률분포)

mathematical function that gives
the probabilities of occurrence of
different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포):
possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포):
possible outcomes have continuous values

Discrete Probability Distribution

- Discrete set of values: $x_1, x_2, x_3, \dots$

Probability of $x_i = p_i \geq 0$

- *e.g.* fair dice: $p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = 1/6$
- Total probability

$$p_{\text{tot}} = \sum_{i=1}^{\infty} p_i$$

Mean Value and Normalization

- Mean value (평균값)

$$\langle x \rangle = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i x_i$$

- Normalization (정규화): $p_{\text{tot}} = \sum_{i=1}^{\infty} p_i = 1$
- Mean value with a normalized distribution

$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i$$

Example 15.2

(1) Find the probability of each of the outcomes of tossing two unloaded dice and (2) find the mean value.

The outcomes range from 2 to 12.

The total number of possible ways is $6 \times 6 = 36$.

$$\langle x \rangle = \sum_{i=2}^{12} p_i x_i$$

$$\langle x \rangle = 7$$

Total (n)	Ways to achieve	Number of ways	Normalized probability = p_n
2	1 and 1	1	$1/36 = 0.02778$
3	1 and 2, 2 and 1	2	$2/36 = 0.05556$
4	1 and 3, 2 and 2, 3 and 1	3	$3/36 = 0.08333$
5	1 and 4, 2 and 3, 3 and 2, 4 and 1	4	$4/36 = 0.11111$
6	1 and 5, 2 and 4, 3 and 3, 4 and 2, 5 and 1	5	$5/36 = 0.13889$
7	1 and 6, 2 and 5, 3 and 4, 4 and 3, 5 and 2, 6 and 1	6	$6/36 = 0.16667$
8	2 and 6, 3 and 5, 4 and 4, 5 and 3, 6 and 2	5	$5/36 = 0.13889$
9	3 and 6, 4 and 5, 5 and 4, 6 and 3	4	$4/36 = 0.11111$
10	4 and 6, 5 and 5, 6 and 4	3	$3/36 = 0.08333$
11	5 and 6, 6 and 5	2	$2/36 = 0.05556$
12	6 and 6	1	$1/36 = 0.02778$

Expected Value (기댓값)

For a function $g(x_i)$ that depends on x_i ,
the expected value is

$$\langle g \rangle = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i g(x_i)$$

e.g. dice game with $g(1) = \text{₩}30,000$

$$\langle g \rangle = \frac{1}{6} \times (\text{₩}30,000) + \frac{1}{6} \times 0 + \cdots + \frac{1}{6} \times 0 = \text{₩}5,000$$

Variance (분산)

$$\sigma_x^2 = \langle (x - \langle x \rangle)^2 \rangle$$

$$\begin{aligned} \sigma_x^2 &= \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i (x_i - \langle x \rangle)^2 = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i (x_i^2 - 2\langle x \rangle x_i + \langle x \rangle^2) \\ &= \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i x_i^2 - \frac{2\langle x \rangle}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i x_i + \cancel{\frac{\langle x \rangle^2}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i} \\ &= \langle x^2 \rangle - 2\langle x \rangle^2 + \langle x \rangle^2 = \langle x^2 \rangle - \langle x \rangle^2 \end{aligned}$$

Example 15.3

Find the variance for the distribution of dice throws in the previous example.

We know: $\langle x \rangle = 7$

Noting that $\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2$,

$$\langle x^2 \rangle = \frac{329}{6}$$

Hence,

$$\sigma_x^2 = \frac{329}{6} - 7^2 = \frac{35}{6}$$

Total (n)	Ways to achieve	Number of ways	Normalized probability = p_n
2	1 and 1	1	$1/36 = 0.02778$
3	1 and 2, 2 and 1	2	$2/36 = 0.05556$
4	1 and 3, 2 and 2, 3 and 1	3	$3/36 = 0.08333$
5	1 and 4, 2 and 3, 3 and 2, 4 and 1	4	$4/36 = 0.11111$
6	1 and 5, 2 and 4, 3 and 3, 4 and 2, 5 and 1	5	$5/36 = 0.13889$
7	1 and 6, 2 and 5, 3 and 4, 4 and 3, 5 and 2, 6 and 1	6	$6/36 = 0.16667$
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9	3 and 6, 4 and 5, 5 and 4, 6 and 3	4	$4/36 = 0.11111$
10	4 and 6, 5 and 5, 6 and 4	3	$3/36 = 0.08333$
11	5 and 6, 6 and 5	2	$2/36 = 0.05556$
12	6 and 6	1	$1/36 = 0.02778$

Standard Deviation (표준편차)

$$\sigma_x = \sqrt{\sigma_x^2} = [\langle x^2 \rangle - \langle x \rangle^2]^{1/2}$$

Continuous Probability Distributions

- Continuous value: x
- Continuous probability distribution (probability density): $f(x)$

$$f(x')dx = \text{Probability of values of } x \in [x', x' + dx]$$

- Probability that $c < x < d = \int_c^d f(x) dx$

Total Probability

$$\int_{x_{\min}}^{x_{\max}} f(x) dx$$

- Normalization

$$\int_{x_{\min}}^{x_{\max}} f(x) dx = 1$$

Mean Value

$$\langle x \rangle = \frac{\int_{x_{\min}}^{x_{\max}} x f(x) dx}{\int_{x_{\min}}^{x_{\max}} f(x) dx}$$

- Mean value with a normalized distribution

$$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x) dx$$

Expected Value

- For a function $g(x)$, the expected value is

$$\langle g \rangle = \frac{\int_{x_{\min}}^{x_{\max}} f(x)g(x) dx}{\int_{x_{\min}}^{x_{\max}} f(x) dx}$$

- e.g. Mean-square value

$$\langle x^2 \rangle = \frac{\int_{x_{\min}}^{x_{\max}} x^2 f(x) dx}{\int_{x_{\min}}^{x_{\max}} f(x) dx}$$

Variance and Standard Deviation

- Variance

$$\sigma_x^2 = \langle (x - \langle x \rangle)^2 \rangle = \frac{\int_{x_{\min}}^{x_{\max}} (x - \langle x \rangle)^2 f(x) dx}{\int_{x_{\min}}^{x_{\max}} f(x) dx}$$

- Standard deviation

$$\sigma_x = \sqrt{\sigma_x^2}$$

Summary

	Discrete	Continuous
Variable	$x_1, x_2, \dots, x_n$	x : real in $[x_{\min}, x_{\max}]$
Probability distribution	p_i	$f(x) dx$
Normalization	$\sum_{i=1}^n p_i = 1$	$\int_{x_{\min}}^{x_{\max}} f(x) dx = 1$
Mean value (normalized dist.)	$\langle x \rangle = \sum_{i=1}^n x_i p_i$	$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x) dx$
Expected value (normalized dist.)	$\langle g \rangle = \sum_{i=1}^n g(x_i) p_i$	$\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x) dx$

Examples and Applications

- Examples

- Binomial probability distribution: discrete
- Uniform probability distribution: continuous
- Gaussian probability distribution: continuous

- Applications

- Quantum mechanics
- Kinetic theory of a gas

Binomial Probability Distribution

(이항확률분포)

Toss an unbiased coin n times, and
find the probability that “heads” will come up m times



Probability Calculation



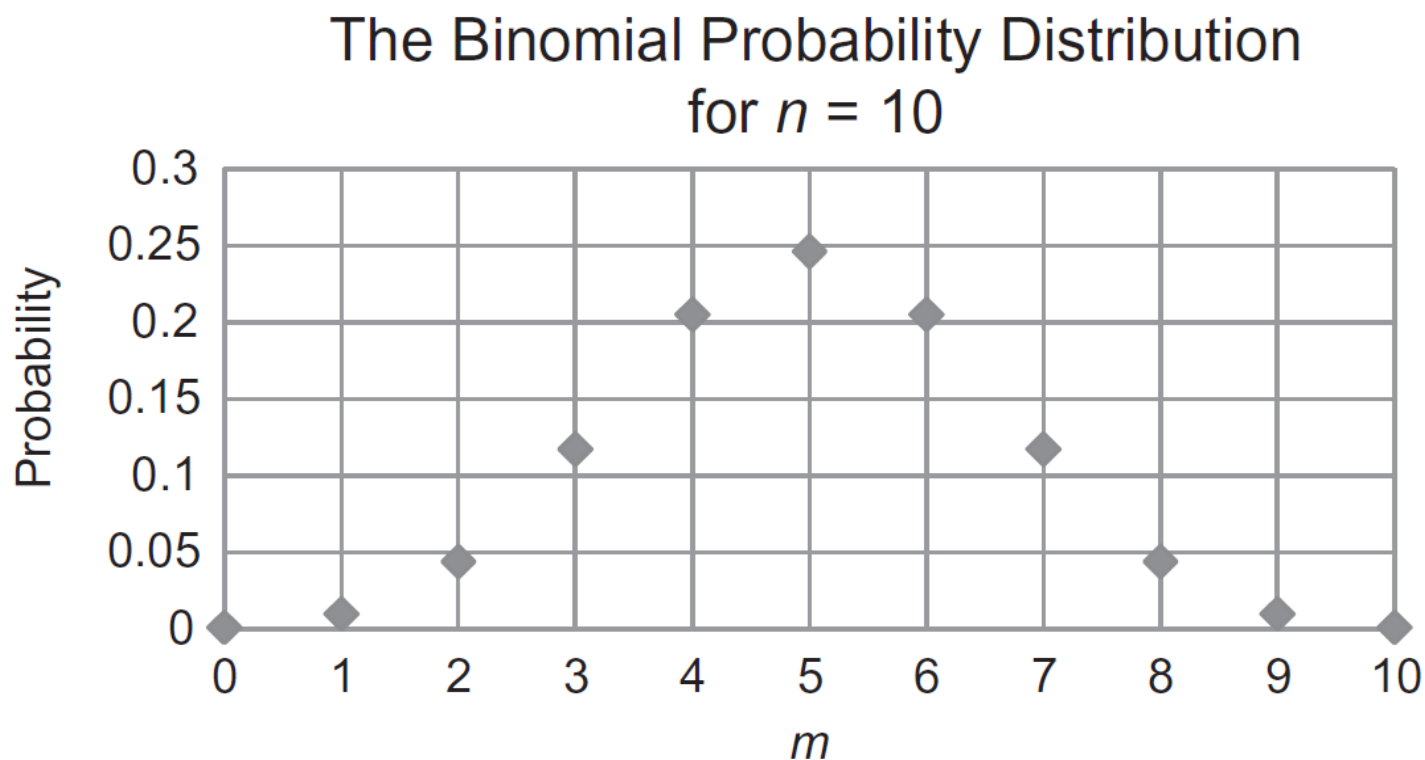
- Number of choosing m coins among n coins

$${}_nC_m = \binom{n}{m} = \frac{n!}{m! (n - m)!}$$

- Probability for a "head" = probability for a "tail" = $1/2$
- Final probability

$$P(n, m) = \frac{n!}{m! (n - m)!} \left(\frac{1}{2}\right)^m \left(\frac{1}{2}\right)^{n-m} = \frac{n!}{m! (n - m)!} \left(\frac{1}{2}\right)^n$$

Plotting the Distribution



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 15.1)

Example 15.4

Calculate the probability that “heads” will come up 50 times if an biased coin is tossed 100 times.

$$p(n, m) = \frac{n!}{m! (n - m)!} \left(\frac{1}{2}\right)^n$$

Here, $n = 100$ and $m = 50$, and

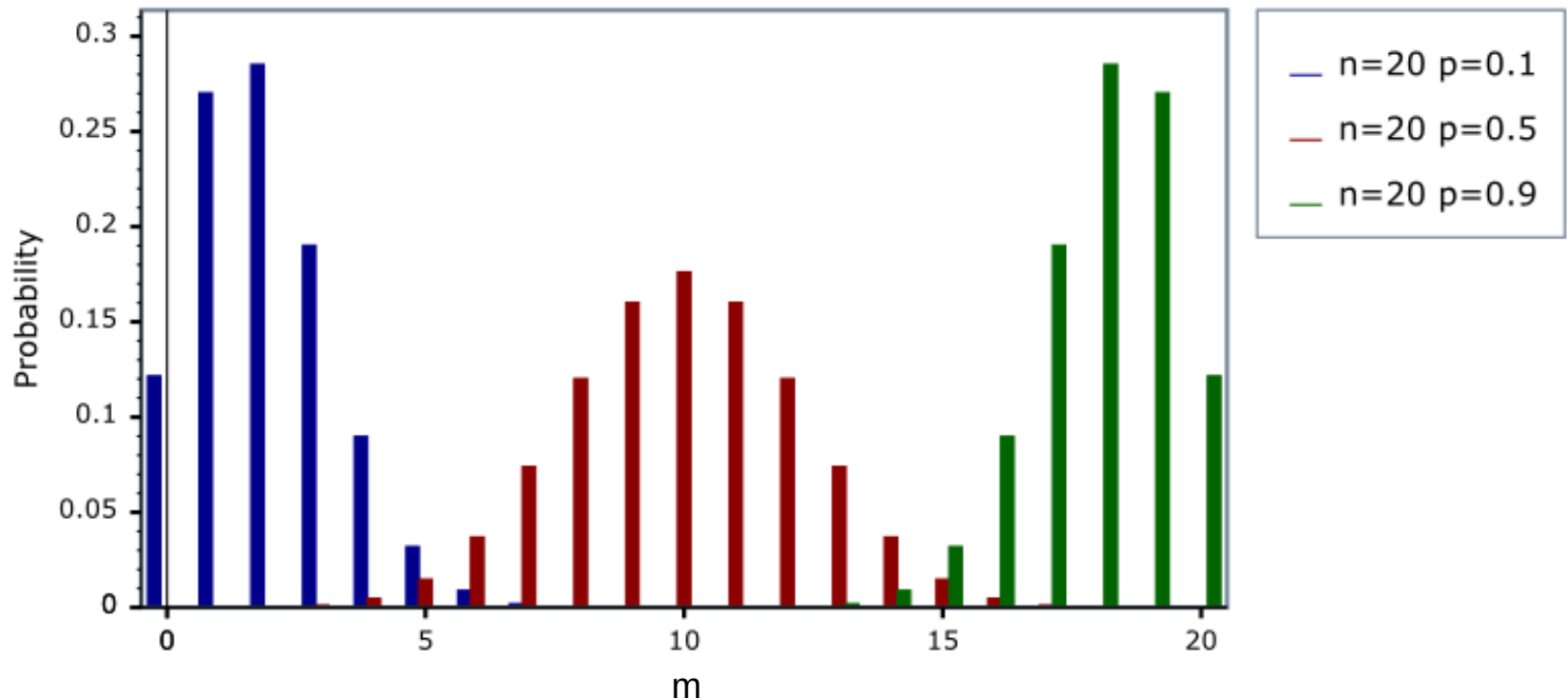
$$p(100, 50) = \frac{100!}{50! 50!} \left(\frac{1}{2}\right)^{100} = 0.07959 \dots$$

With Biased Coins

- p = probability that “heads” will occur
- $(1 - p)$ = probability that “tails” will occur

$$P(n, m) = \frac{n!}{m! (n - m)!} p^m (1 - p)^{n-m}$$

Binomial Probability Distributions



Mean Value and Variance

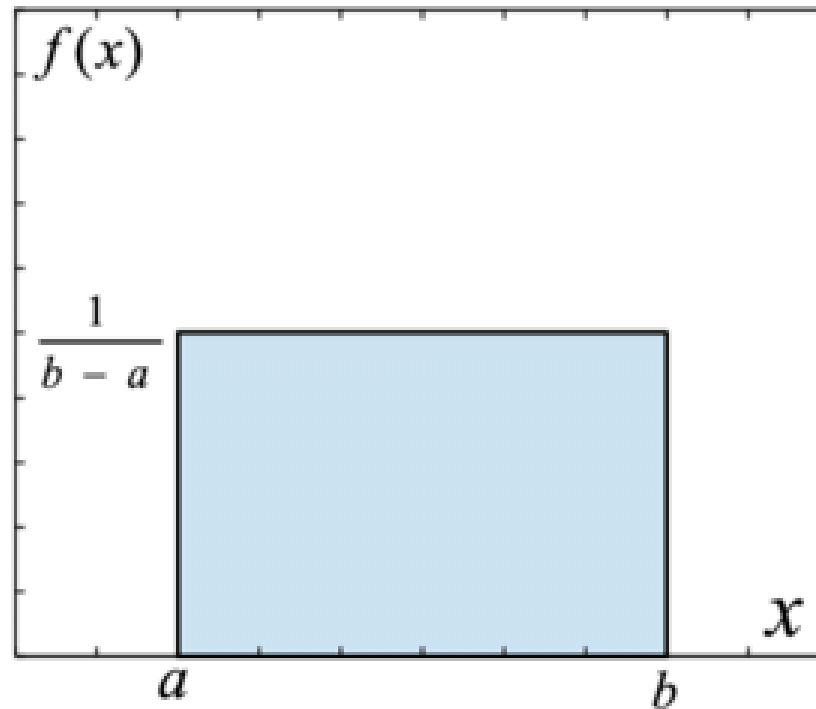
- Mean value

$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i = \sum_{m=0}^n \frac{n!}{m! (n-m)!} p^m (1-p)^{n-m} \times m$$
$$\langle x \rangle = np$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = np(1-p)$$

Uniform Probability Distribution



Mean Value and Mean-Square Value

- Mean value

$$\langle x \rangle = \int_a^b x f(x) dx = \int_a^b \frac{x}{b-a} dx = \frac{1}{b-a} \left[\frac{1}{2} x^2 \right]_a^b = \frac{a+b}{2}$$

- Mean-square value

$$\begin{aligned} \langle x^2 \rangle &= \int_a^b x^2 f(x) dx = \int_a^b \frac{x^2}{b-a} dx = \frac{1}{b-a} \left[\frac{1}{3} x^3 \right]_a^b \\ &= \frac{a^2 + ab + b^2}{3} \end{aligned}$$

Variance

$$\langle x \rangle = \frac{a+b}{2}; \quad \langle x^2 \rangle = \frac{a^2 + ab + b^2}{3}$$

$$\begin{aligned} \sigma_x^2 &= \langle x^2 \rangle - \langle x \rangle^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{a^2 - 2ab + b^2}{12} = \frac{(a-b)^2}{12} \end{aligned}$$

Example 15.5

Find the mean, the variance, and the standard deviation for a uniform probability distribution with $a = 0$ and $b = 10$. Find the probability that x lies between $\langle x \rangle - \sigma_x$ and $\langle x \rangle + \sigma_x$.

$$\langle x \rangle = \frac{a + b}{2} = \frac{0 + 10}{2} = 5$$

$$\sigma_x^2 = \frac{(a - b)^2}{12} = \frac{(0 + 10)^2}{12} = \frac{25}{3}$$

$$\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}}$$

$$\text{Probability} = \int_{\langle x \rangle - \sigma_x}^{\langle x \rangle + \sigma_x} f(x) dx = \int_{5 - 5/\sqrt{3}}^{5 + 5/\sqrt{3}} \frac{1}{10 - 0} dx = \frac{1}{\sqrt{3}}$$